

Synopsis Presentation on **UNIVERSAL AI AUTO-FILL ASSISTANT**

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1. Introduction

In today's digital world, users interact with many online platforms daily.

Most of these platforms require users to fill forms with the same personal information repeatedly.

This repetitive process wastes time and reduces productivity.

Real-World Scenarios

- Job Seekers: ~15 hours/week spent filling applications repeatedly
 - Doctors: Manual patient forms lead to errors and burnout
 - Students: Re-enter personal data for visas, registrations, banking
- Global Impact
- Manual form filling wastes 47 billion hours every year



Why Current Solutions Fail

Password Managers

- Fill login fields only
- Cloud storage raises privacy concerns

Google Autofill

- Around 23% app incompatibility

Browser Extensions

- Work only in browsers
- Fail inside native mobile applications

The Core Gap

No existing solution provides all three:

- Universal compatibility across applications
- Intelligent field recognition using AI
- Complete privacy through on-device processing

2. Literature Survey

Year	Authors	Title	Key Finding
2024	Fu et al	Understanding Mobile GUI: From Pixel-Words to Screen Sentences	Proposed a pixel-based UI understanding framework that extracts text and visual components from screenshots to understand mobile interfaces for automation tasks
2023	Chen et al	Deep Learning for Mobile Form Field Classification	94% accuracy in field prediction using CNNs on UI hierarchies
2023	Song et al.	VisionTasker: Mobile Task Automation Using Vision-Based UI Understanding	Developed a framework that combines UI vision analysis and AI planning to automate tasks on mobile apps by interpreting screen content
2022	Rodriguez &Kumar	Accessibility-Based Text Injection in Android Applications	Demonstrated bypassing from restrictions via accessibility APIs
2021	Patel et al	On-Device NLP for Context-Aware Form Filling	Lightweight BERT models for field understanding

2.2 Problem Statement

Develop a privacy-first Android autofill system achieving:

Metric	Target
Application Compatibility	$\geq 98\%$ of Android apps (including restricted apps)
Field Recognition Accuracy	$\geq 95\%$ across diverse form layouts
Supported Field Types	50+ form field categories
Form Filling Time	≤ 2 seconds
Privacy	100% on-device processing (no cloud transmission)
OCR Accuracy	$\geq 90\%$ extraction from physical documents
Battery Consumption	$\leq 3\%$ increase during active use

3. Proposed Solution Overview

Stage 1 – Profile Creation

- Create user profiles (Personal / Work / Student)
- Data entered manually or via OCR scan



Stage 2 – Screen Analysis

- Accessibility service detects form fields
- ML model performs field classification



Stage 3 – Smart Profile Selection

- System analyzes app context
- Automatically selects correct profile



Stage 4 – Dual-Mode Filling

- Standard Mode: Android Autofill framework
- Force-Fill Mode: AI bubble for restricted apps



Result: Complete form filling in < 2 seconds



4. Objectives

- Build a smart autofill system that can automatically fill forms in most Android apps (including restricted apps) with high compatibility and fast performance.
- Develop a lightweight machine learning model that can identify different types of form fields (50+ types) with high accuracy to improve automatic form filling.
- Ensure secure user data management by using encrypted storage, biometric authentication, and an OCR scanner to extract information from documents like ID cards and business cards.

5. Contributions

Technical Contributions:

- Hybrid Filling Architecture: First dual-mode system (Standard + Force-Fill)
- Lightweight ML Model: 12MB size, 50ms inference, 95% accuracy
- Context-Aware Selection: Auto profile switching based on app
- Privacy Architecture: Zero data transmission

Academic Contributions:

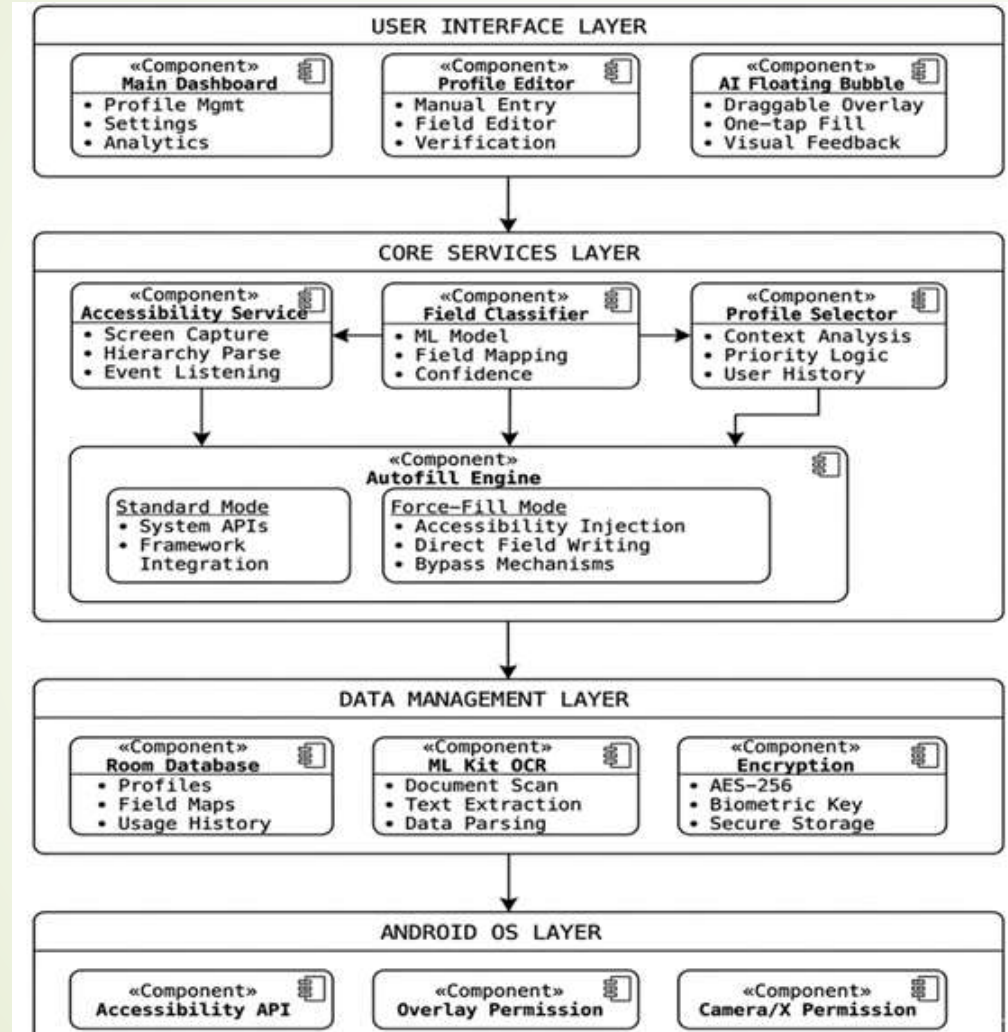
- Compatibility study across 200+ Android apps
- Dataset of 10,000+ labeled form fields

UX Contributions:

- Zero-learning floating bubble interface
- Transparent privacy controls

6. Theoretical Analysis

6.1 Block Diagram





6.2 Hardware / Software Requirements

Software Requirements:

- Kotlin, Jetpack Compose
- Room DB + SQLCipher (Encryption)
- Android Accessibility
- Android 8.0+ (API 26)

Hardware Requirements:

- 2GB RAM minimum
- 50MB storage minimum
- Camera



7. Applications

- **Employment:** LinkedIn, Naukri, company portals
- **E-commerce:** Amazon, Flipkart checkout
- **Healthcare:** Patient registration, telemedicine
- **Education:** College apps, course registration
- **Finance:** Bank accounts, loan applications
- **Government:** Visa forms, tax filing
- **Travel:** Flight, hotel bookings
- **Accessibility:** Assistive tech for motor challenges



GitHub Link

Pushpraj Singhal : <https://github.com/pushprajsinghal>

Samyak Choudhary : <https://github.com/Upriser99>

Vinay Kumar Patel : <https://github.com/Vinaypatel24>

Devesh Talreja : <https://github.com/devesh-talreja>

8. REFERENCES

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Thank You

*Thank you for your time and attention.
Questions are welcome.*

Queries ?